Causal Claims in Economics

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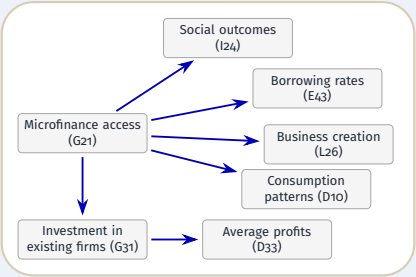
Project: **causal.claims**

Thiemo Fetzer
Warwick / Bonn / CEPR



Example claim graph Banerjee et al. (2015)

Economics has a representation problem

The bottleneck has moved.

It is now cheaper to produce papers than to organize what they actually claim.

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3. Synthesis stays manual

If cumulative science matters, prose alone is a poor storage format for comparison and reuse.

Why prose is the wrong storage layer

What the literature gives us

Question: often implicit, spread across abstract and introduction.

Mechanism: buried in chains of sentences and section-specific language.

Evidence: mixed together with rhetoric, tables, and design discussion.

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Nodes: standardized economic concepts.

Edges: explicit source to sink relationships.

Evidence labels: causal design, non-causal support, tentativeness, novelty, and position.

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The contribution is not just a PDF.

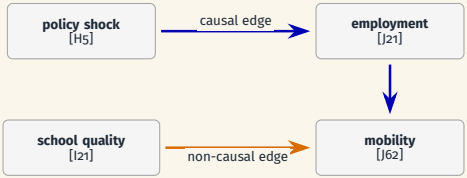
It is a linked claim structure that can be compared across papers.

Claim graphs

A paper as a graph

What the object looks like before the pipeline.

A claim graph is a paper-level object



Nodes

Economic concepts mapped to a common ontology: here, JEL codes.

This is the core move: represent argumentative structure in a comparable, aggregable form.

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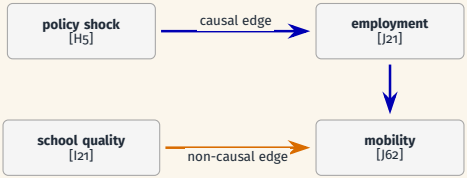

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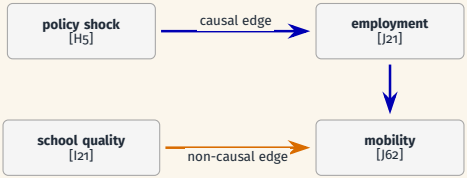

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Evidence annotation

Each edge stores whether support comes from DiD, IV, RCT, RDD, synthetic control, or non-causal evidence.

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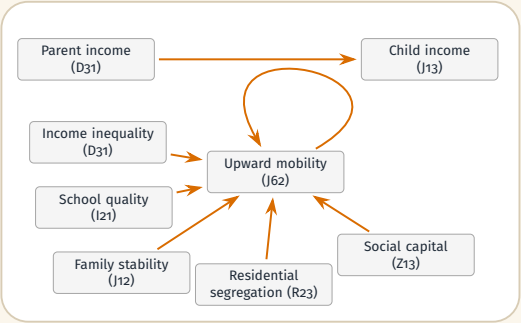
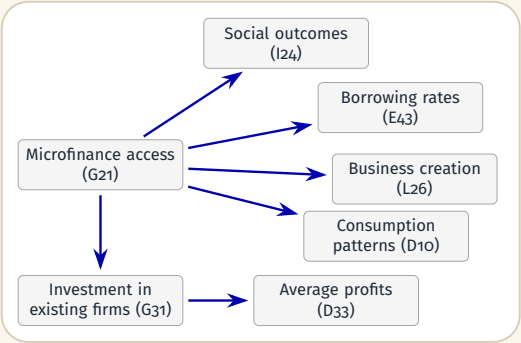
Before the pipeline, make it concrete



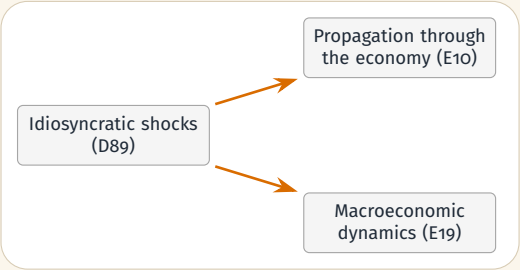
Chetty et al. (2014) Broad correlational map around upward mobility



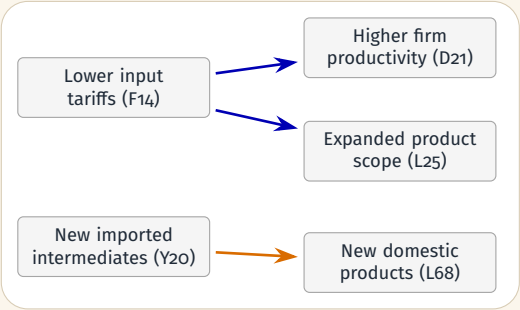
Banerjee et al. (2015) Compact causal chain with RCT-backed outcomes

Same discipline, same status, very different claim architecture.

Two more landmark graphs



Gabaix (2011) Compact non-causal mechanism structure



Goldberg et al. (2010) Mixed causal and non-causal claim structure

The object captures different architectures, not just different topics.

What the object captures – and what it does not

What claim graphs capture

✓ Which concepts a paper connects
✓ Whether support is causal or non-causal
✓ Complexity, novelty, and topic position

✓ Paper-level structure that can be compared at scale

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✗ They do not decide whether a claim is true
✗ They do not recover every nuance of variable wording

✗ They do not eliminate the need for validation and judgment

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Interpretation rule

This is measurement infrastructure. It extracts what papers *state* and how they support it. That is already a large step forward.

Measurement

Build the infrastructure

How the retrieval pipeline turns papers into claim graphs.

Corpus, scope, and design choices

44,852

NBER + CEPR working papers

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1980–2023

Four decades of economics

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Publication + citation snapshot

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Why the first 30 pages?

In economics, the abstract, introduction, identification strategy, and main results usually contain the paper's central claims. A fixed cap keeps decades comparable as papers lengthen.

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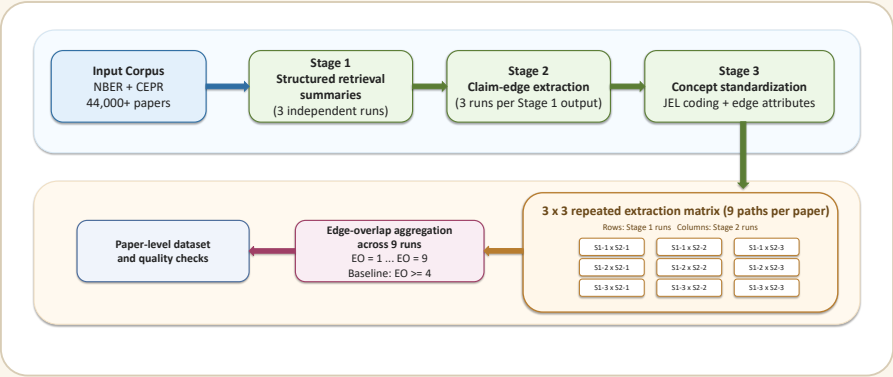
What is intentionally narrow

Causal edges are defined conservatively: DiD, IV, RCT, RDD, event-study-style DiD/TWFE implementations, and synthetic control.

Extraction pipeline: retrieve, standardize, aggregate



Stage 1: qualitative curation

Three first-30-page summaries recover the question, design cues, and claim snippets.

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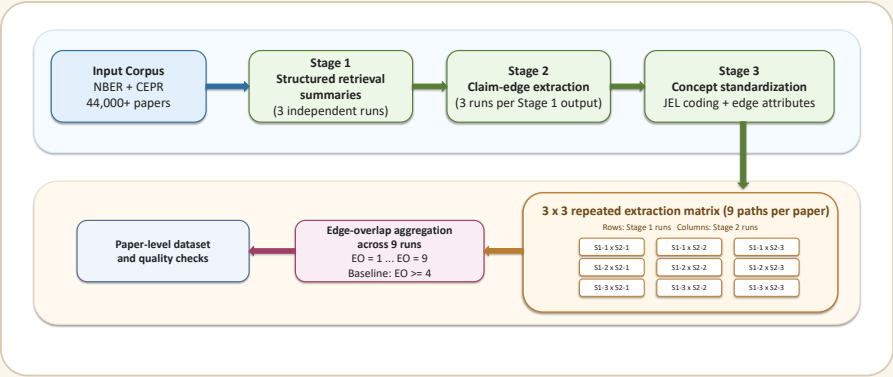

Stage 2: graph extraction

Convert each summary into labeled source-to-sink edges.

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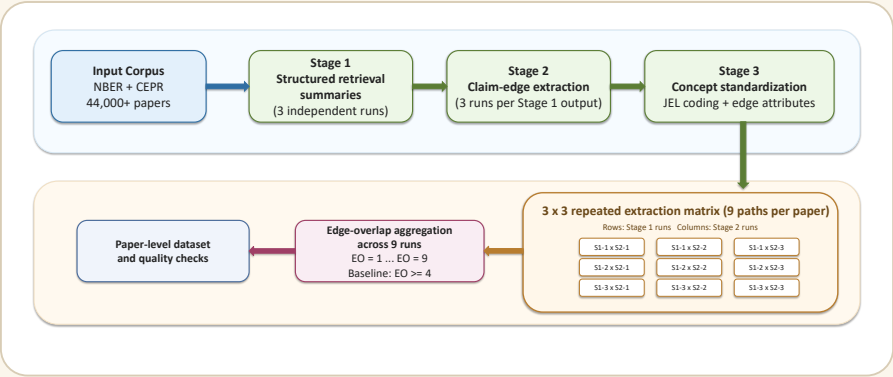
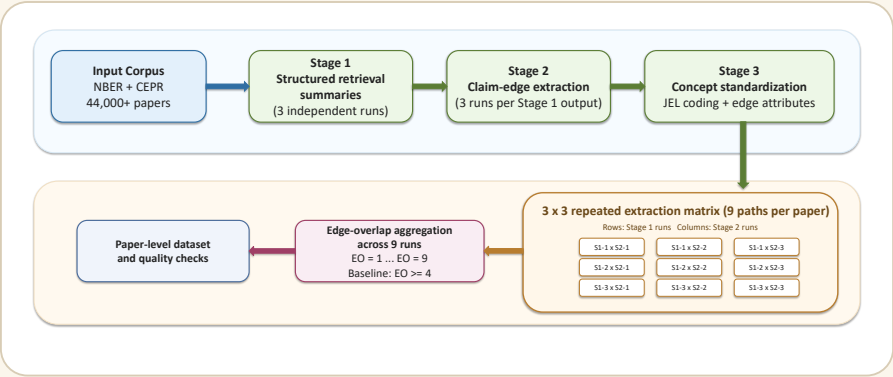


Stage 3: concept standardization

Map endpoints to JEL concepts.

Extraction pipeline: retrieve, standardize, aggregate



Stage 4: EO aggregation

Retain recurring edges; EO >= 4 is the baseline.

Edge-overlap makes precision and recall explicit

Why this matters

A single LLM pass is a noisy measurement. Repeating the same constrained extraction and retaining overlapping edges makes the trade-off transparent rather than mystical.

Interpretation

Lower EO keeps more candidate edges but accepts more noise. Higher EO suppresses fragile edges but thins the graph. The paper uses EO >= 4 as the operating point.

Why trust the measurement at all?

External benchmark 1

Match method and field labels against the Brodeur et al. dataset of leading economics papers.

External benchmark 2

Compare causes, effects, and exogenous variation with the Plausibly Exogenous Galore dataset.

Internal audit

Re-extract edges from verbatim snippets only and inspect the precision–recall trade-off across EO thresholds.

Bottom line

Treat this as a measurement system, not an oracle. The paper validates the retrieval on method labels, concept alignment, exogenous variation, and stability under stricter overlap thresholds.

More details: Brodeur | Plausibly Exogenous | Snippet / EO audit

What does the measurement show?

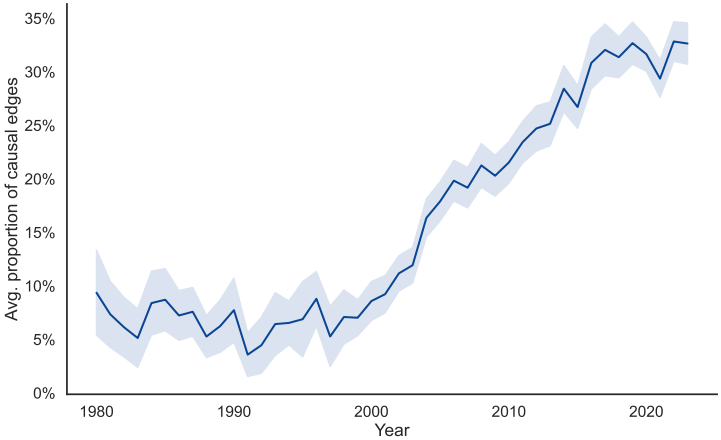
Start with how causal evidence changed across economics.

Descriptive results

What changed in economics?

The claim-level view of the credibility revolution.

The share of causal edges rises sharply



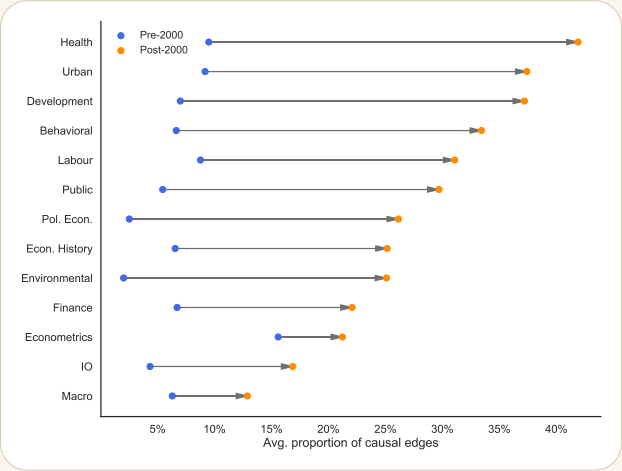
First headline

7.7% in 1990
to
31.7% in 2020

Causal support becomes common rather than exceptional in the paper-level claim graph.

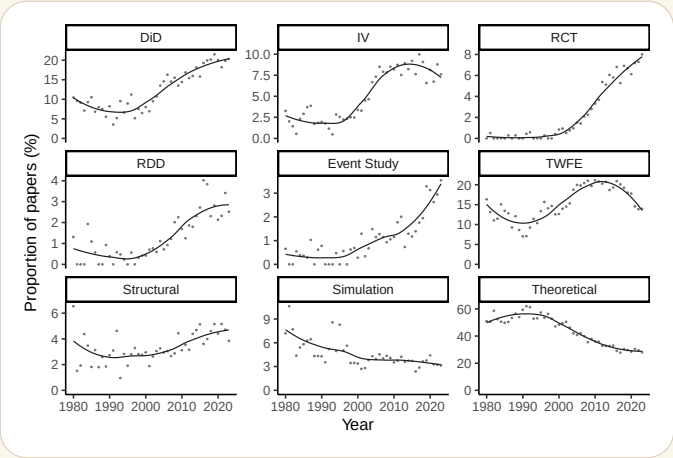
Why this matters: the credibility revolution is visible not just in methods used by papers, but in the share of relationships backed by causal designs.

The rise is broad – but not uniform across fields



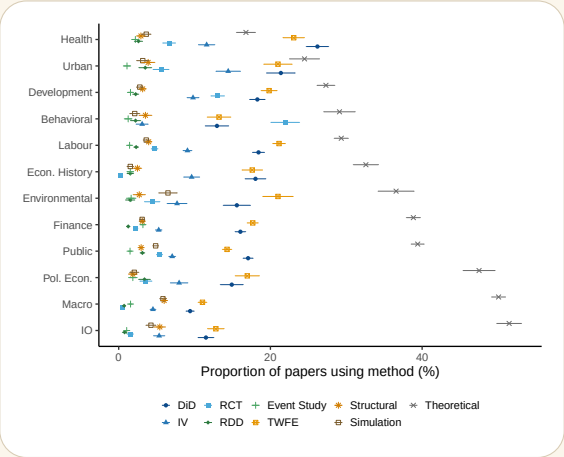
The diffusion is discipline-wide, but still shaped by field-specific data environments and identification traditions.

Method diffusion confirms the same story



DiD, IV, and RCT rise, while theory-heavy papers become a smaller share of the working-paper universe.

Fields still have very different method cultures



Method culture remains field-specific: DiD is strongest in urban/labour/public, RCTs in behavioral/development, and theory remains concentrated in macro, IO, and finance.

Incentives

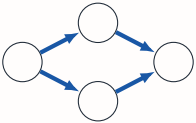
What kinds of claim structures get rewarded?

Incentives

What gets rewarded

Publication and citation patterns.

Three families of paper-level measures



Complexity

How broad and deep is the argument?

Edges, unique paths, longest path

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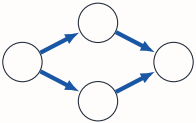

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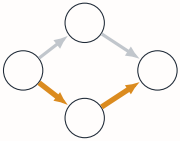

Novelty

Does the paper add genuinely new edges, paths, or gap-filling combinations?

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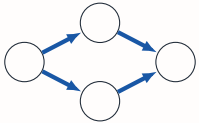

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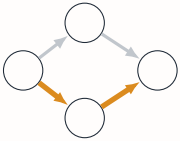

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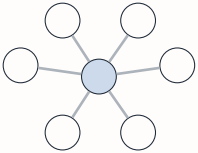
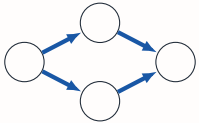


Position

Are the paper's concepts central, peripheral, or mixed inside the cumulative literature graph?

Three families of paper-level measures



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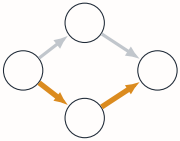

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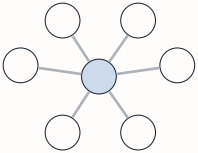

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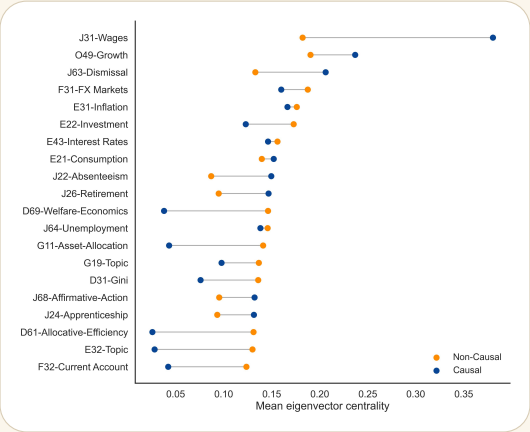
Main comparison

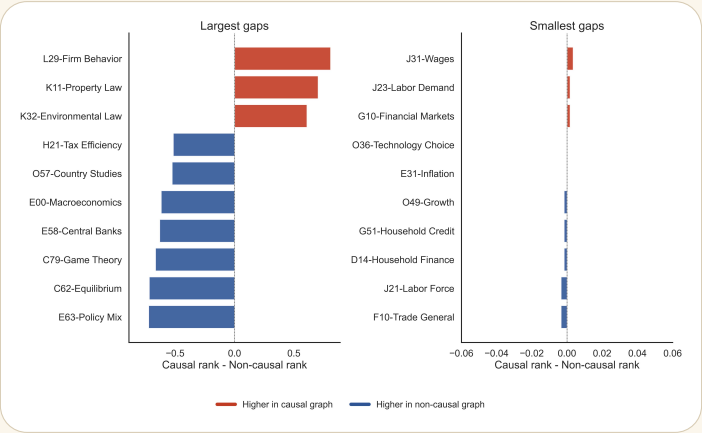
Every family is computed in both causal and non-causal variants. That split is the paper's main empirical lever.

Which concepts stay central once we focus on causal edges?



Blue is causal centrality and orange is non-causal centrality. The x-axis is mean eigenvector centrality, not a regression coefficient.

Concept centrality shifts once we focus on causal edges



Some concepts move sharply up or down once the literature graph is restricted to causally identified edges.

The empirical question is deliberately simple

Outcome 1

Top journal placement

Top 5, 6–20, and 21–100 publication outcomes

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Outcome 2

Long-run influence

Log citations as of the September 2024 snapshot

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Associational setup

Regress each outcome on one graph measure with year fixed effects.

Then compare causal and non-causal versions of the same measure family.

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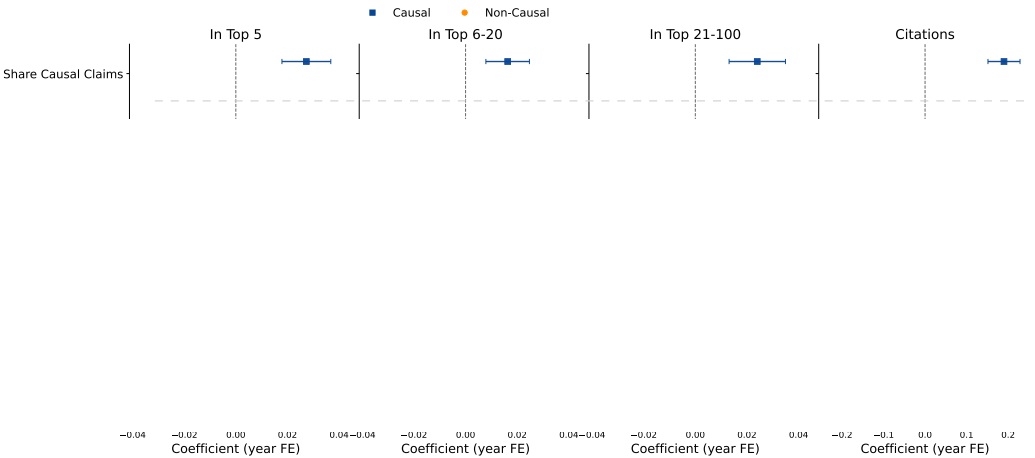
Hypothesis

Papers with richer causal structure and novelty should outperform papers whose extra structure is mostly non-causal.

Build the predictors result, one family at a time



1. Causal share Higher causal share improves placement and citations.

Variant measures

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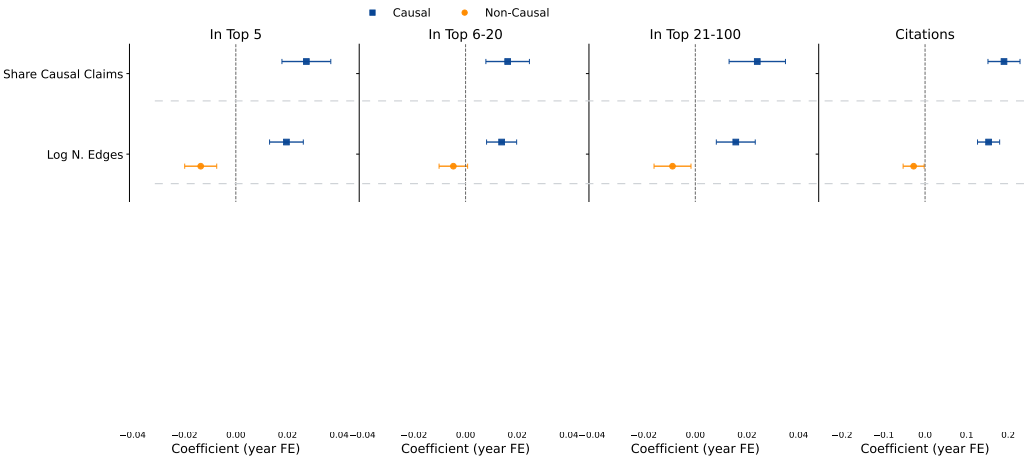

2. Complexity More causal edges help; the non-causal analogue does not.

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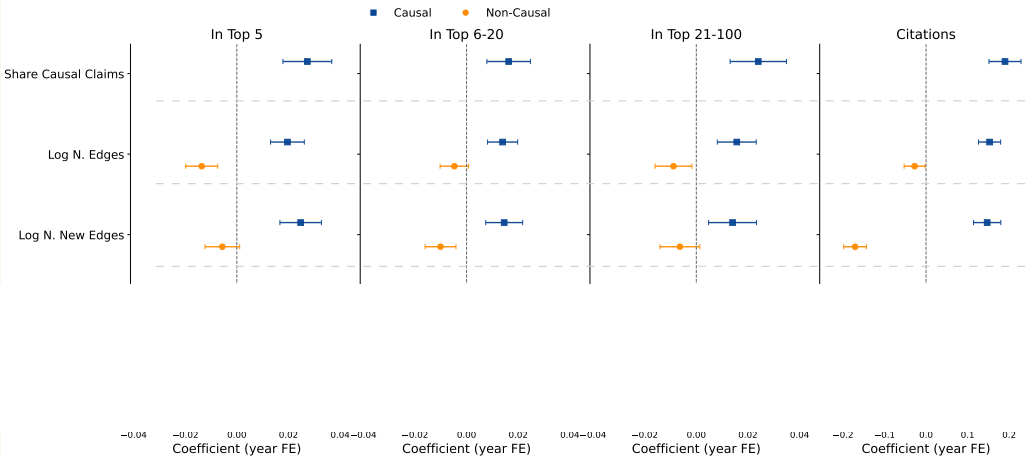

3. Novelty New causal edges predict payoff; new non-causal edges do not.

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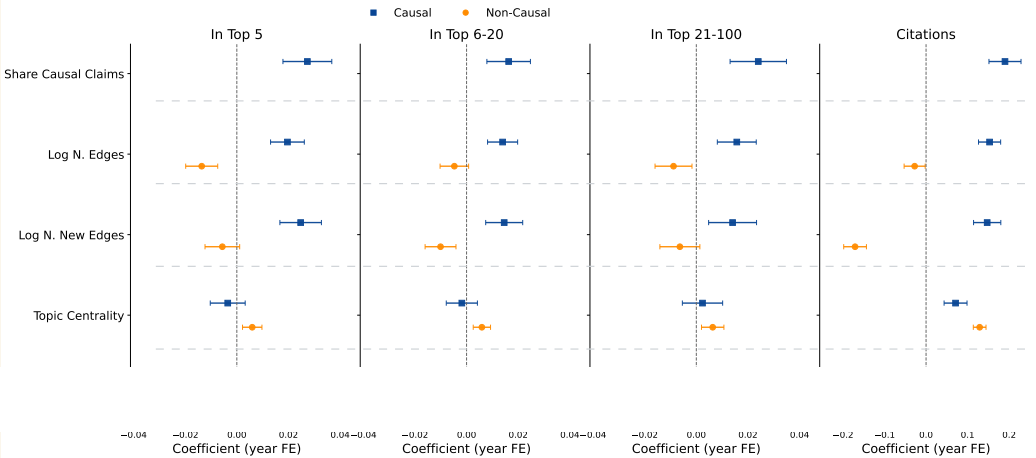
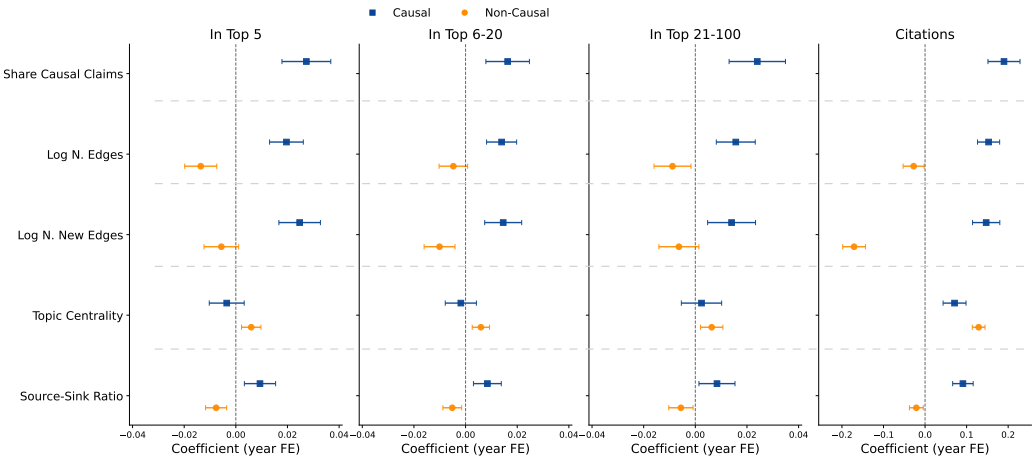


4. Centrality Centrality helps citations more than editorial selection.

Variant measures

Build the predictors result, one family at a time



5. Source-heavy Source-heavy causal structure is rewarded; non-causal is not.

Variant measures

Selection and diffusion are not the same game

What top journals reward

causal depth
causal novelty
richer causal pathways

identified structure that looks frontier-like and editorially bold

vs.

What citations still like

conceptual centrality
recognizable conversations

broad diffusion is helped by sitting close enough to the discipline's main graph

Practical implication: the strategies that maximize elite placement are not identical to the strategies that maximize long-run attention. The profession rewards causal innovation, but the citation market still likes conceptual embeddedness.

What this still does not do

1. Truth

No truth test.

Records stated claims, not correctness.

2. Nuance

Some nuance is lost.

Mapping and extraction compress narrative detail.

3. Scope

The window is fixed.

The 30-page window can miss later detail.

Still: even with conservative scope and explicit validation, the object is already rich enough to recover field trends, method diffusion, and the reward structure around causal narrative depth and novelty.

Closing

Claim graphs turn papers into queryable objects.

That makes literatures easier to compare, aggregate, and extend.

causal.claims

Appendix

Robustness, validation, and extra visuals

Optional slides for discussion and follow-up questions.

EO robustness: the causal-edge trend is not a threshold artifact

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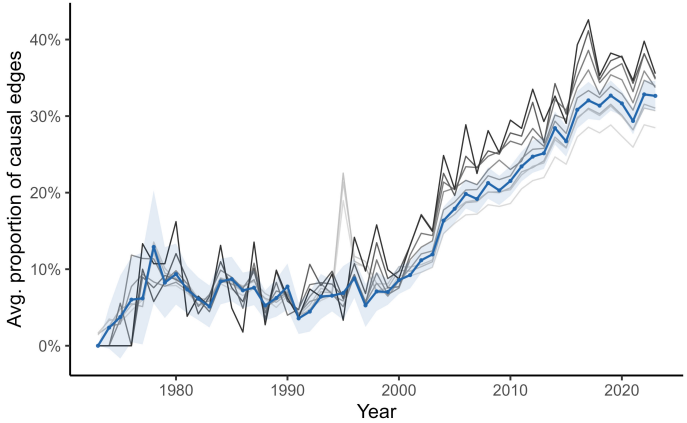
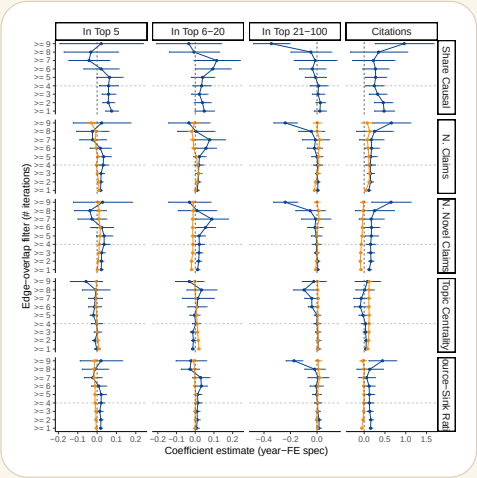


EO robustness: headline regression signs remain stable

Back



Takeaway

Signs are stable through EO >= 4.

Intervals widen only once support becomes very thin.

Validation: Brodeur benchmark

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Matched sample sizes

EO >= 4: 285 papers EO >= 7: 169 papers EO = 9: 90 papers

Method recovery

Explicit quasi-experimental designs are recovered well in matched papers, especially for IV and for the cleaner DiD and RDD cases.

Threshold pattern

Raising the EO threshold improves precision, but support thins out as we keep only the most strongly backed edges.

Why this matters

This external benchmark suggests the pipeline is not just finding economics words; it is recovering recognizable causal-design labels.

The validation target here is recognizable method labeling, not full causal adjudication.

Validation: Plausibly Exogenous benchmark

Back

Matched sample sizes

EO >= 4: 491 papers EO >= 7: 296 papers EO = 9: 148 papers

Semantic alignment

Retrieved snippets line up meaningfully with the cause, effect, and exogenous-variation language in the benchmark corpus.

Strongest signal

Exogenous variation is the cleanest component across thresholds, which is exactly the part we most want to isolate in causal claims.

Why this matters

The pipeline is not only tagging designs; it is also finding the semantic content around the shock or variation that identifies the claim.

The strongest semantic lift comes from text describing exogenous variation.

Validation: snippet / EO audit

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Audit scale

77,070 snippet-level checks covering 864,085 supported edges in the lowest-overlap setting.

Precision-recall trade-off

As the EO filter tightens, precision rises sharply while recall falls, which is the expected behavior of a conservative overlap rule.

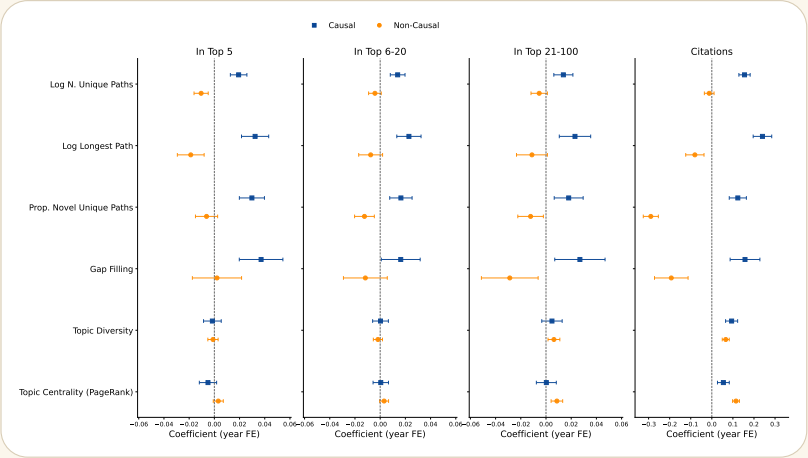
Operational takeaway

EO is doing real screening work: stricter settings give cleaner evidence at the cost of keeping fewer supported edges.

This audit is about extraction quality control: cleaner support comes from stricter overlap, not from changing the claim definition itself.

Predictor variants tell the same directional story

Back



Variant measures for depth, path novelty, and topic diversity continue to favor causal over non-causal structure in top-tier placement and citations.

Gap filling: top journals reward underexplored combinations unevenly

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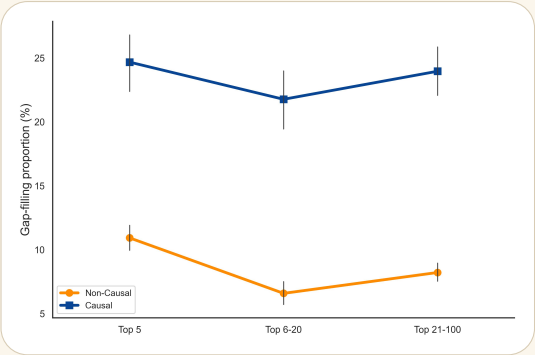

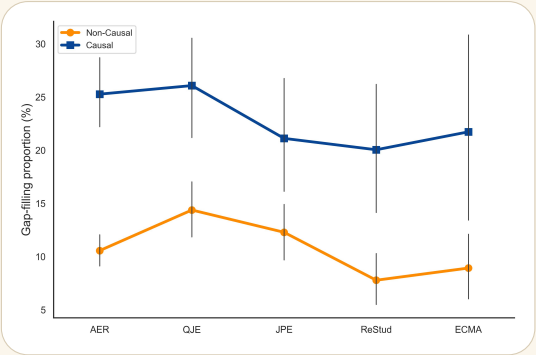

Causal gap filling appears especially strong in journals like QJE and AER, but the citation payoff is less systematic than for causal depth or causal novelty.

Validation summary in one slide

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Brodeur benchmark

Strong label recovery for several methods and fields; especially good performance for RDD, Macro, and Urban labels in matched papers.

Plausibly exogenous benchmark

High maximum semantic alignment for causes, effects, and especially sources of exogenous variation.

Snippet audit

Precision rises from roughly 0.78 to 0.94 as overlap thresholds tighten, with the expected recall trade-off.

Together, the three checks cover label recovery, semantic alignment for exogenous variation, and precision–recall trade-offs inside the extraction pipeline.

Data and replication notes

What is released

Derived edge lists, paper-level measures, prompts, schemas, and analysis code via the project website.

What is not released

Proprietary source PDFs are not redistributed, but the package supports regeneration once users re-download NBER and CEPR source documents.

Why that matters

The contribution is both a substantive finding and a reusable claim-level infrastructure for future meta-research.